EXPERIMENT 1

BASIC NETWORK TOPOLOGIES USING SWITCHES AND HUBS

Course:

ENCS304 – Computer Networks

Name:

Khushi Yadav

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1. Objective

To design and analyze Star, Hub-based and Ring-like topologies using Cisco Packet Tracer and evaluate their connectivity, performance, and fault tolerance.

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2. Tools Used

- Cisco Packet Tracer
- Switch (2960-24TT)
- Hub
- PCs
- Copper Straight-through and Cross-over cables

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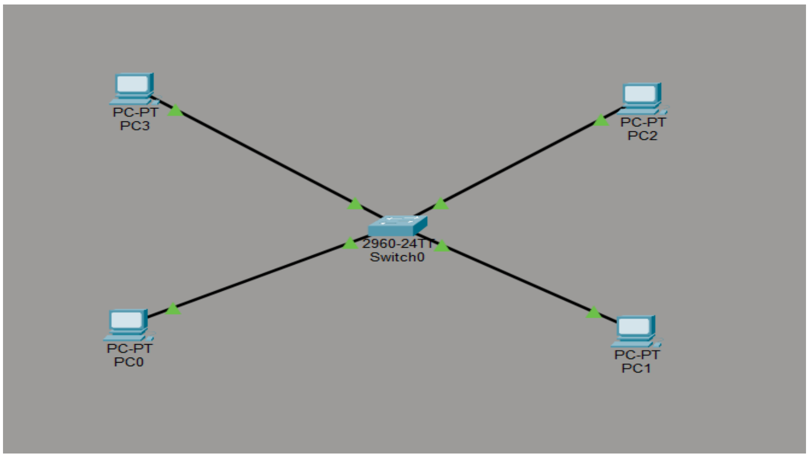
3. Implementation and Results

3.1 Star Topology (Switch Based)



Four PCs were connected to a central switch.

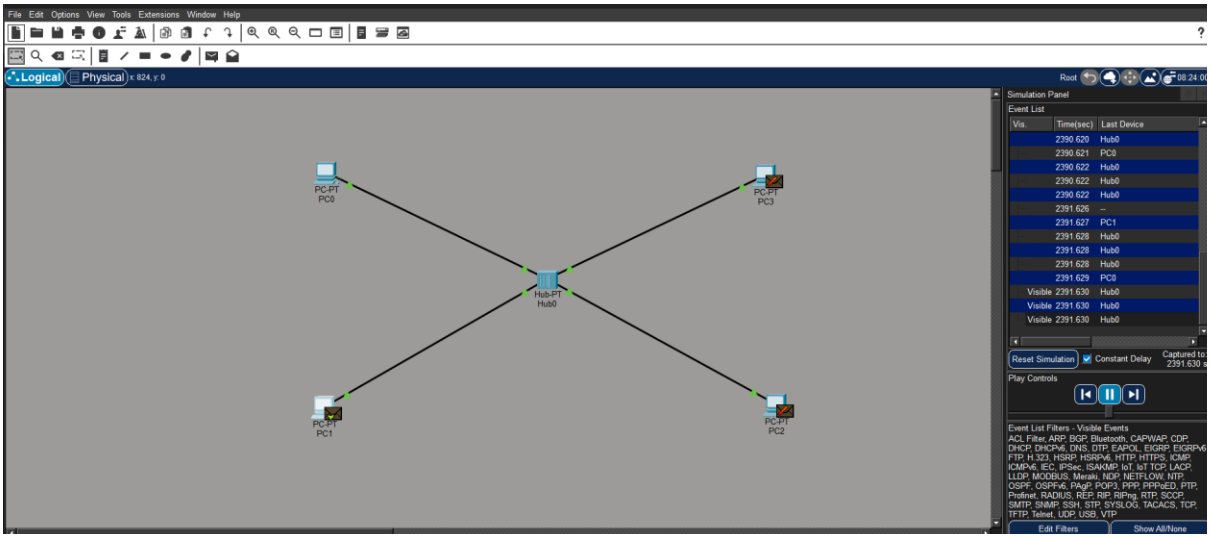
Result:

All ping requests were successful with 0% packet loss.

Observation:

The switch forwarded data only to the intended destination port. No collisions were observed. Performance was efficient.

3.2 Hub-Based Topology



Four PCs were connected to a hub.

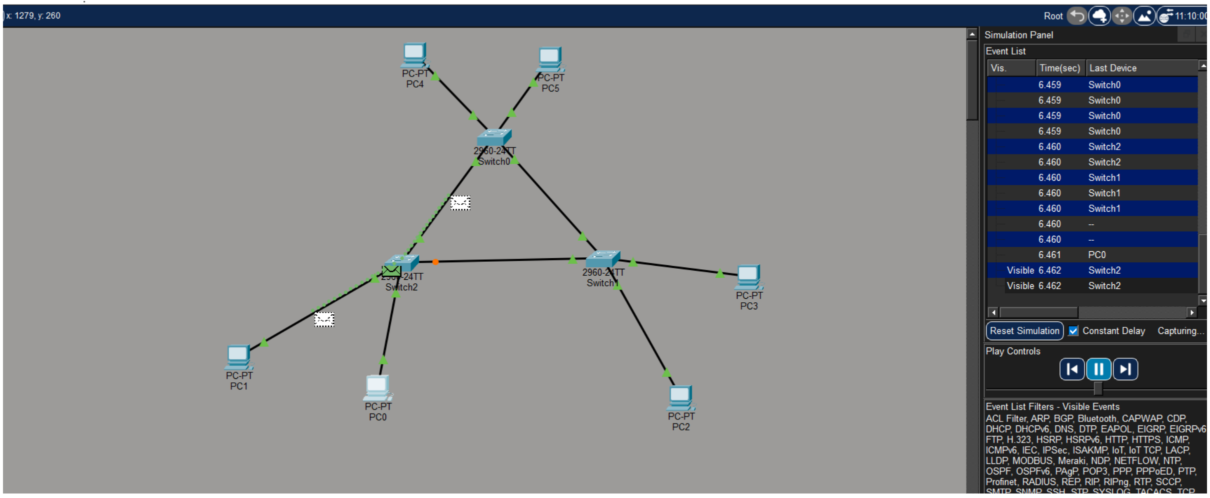
Result:

All pings were successful.

Observation:

Packets were broadcast to all ports.
Collisions were observed during simultaneous ping.
Performance was slower compared to switch.

3.3 Ring-like Topology (Loop using Switches)



Three switches were connected in loop form.

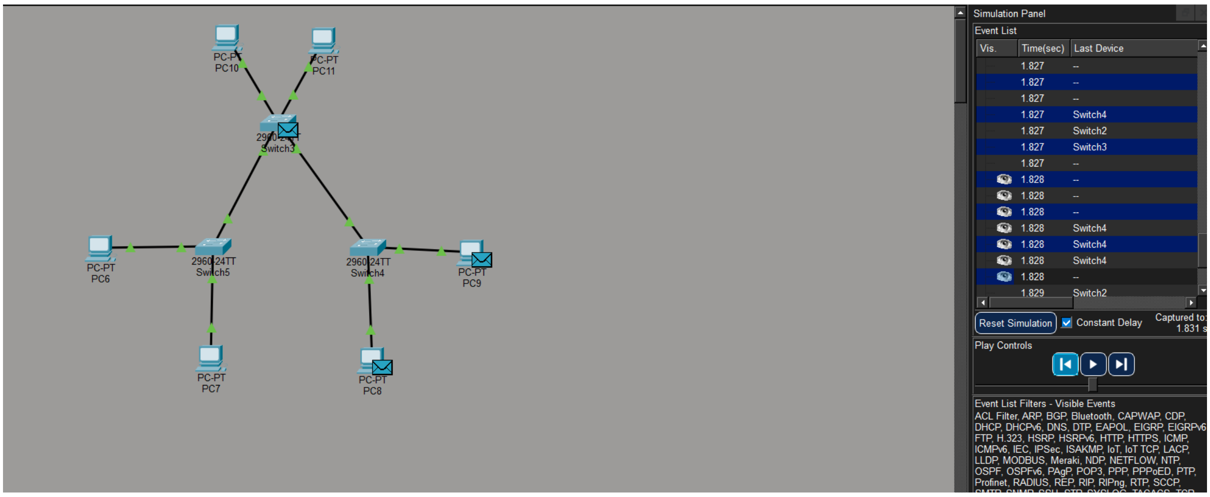
Result:

Ping across switches was successful.

Observation:

Spanning Tree Protocol blocked one redundant link (orange state).
Packets followed active forwarding path.

3.4 Failure Test



One switch link was disconnected.

Result:

Communication continued successfully.

Observation:

STP re-enabled alternate path.
Network showed fault tolerance capability.

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4. Comparison

Topology	Performance	Collision	Fault Tolerance
Star	High	No	Medium
Hub	Low	Yes	Low
Ring	Medium	No	High

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5. Conclusion

This experiment demonstrated the behavior of different network topologies. Switch-based topology provides better performance and collision-free communication. Hub-based topology suffers from collisions due to broadcast behavior. Ring topology offers redundancy and fault tolerance through Spanning Tree Protocol.

The experiment provided practical understanding of LAN design and physical network planning.